When AI Agents Disagree

UCSC AI Hackathon Challenge October 2026

From Collaboration to Collective Intelligence

Challenge Sponsor: Cisco Outshift | AGNTCY

The Challenge

Can you build AI agents that work together, adapt to changing conditions, and make decisions even when they disagree?

Individual agents can reason, use tools, and solve complex problems. This challenge asks what happens when independent agents bring different capabilities, incomplete information, competing priorities, or conflicting recommendations to the same problem.

Choose one of three levels: collaboration, adaptation, or collective decision-making. Participants can build on AGNTCY and its coffeeAGNTCY reference application, or use any other agent harness or multi-agent system (MAS) framework they are familiar with. No specific framework, AI model, or platform is required.

Level 1 Build an AI Team

Mission Build or extend a multi-agent system with specialized agents that collaborate to solve a problem.

Introduce agents with distinct capabilities. Show them exchanging information and coordinating actions to complete a task that needs contributions from multiple areas of expertise. Extend the starting application with meaningful new behavior.

Example projects Add shipping, quality control, or sustainability agents to a coffee supply chain; build a campus assistant for transportation, scheduling, and events; or create a research team for gathering, analyzing, and verifying information.

Success criteria Demonstrate distinct capabilities, meaningful communication, and a completed shared task. Explain why multiple agents are useful and identify what your team added beyond the starting framework or demo.

What can your team of AI agents accomplish together?

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Level 2 Build an Adaptive AI Team

Mission Create agents that dynamically discover collaborators, adapt to changing conditions, and resolve conflicting recommendations.

Move beyond a fixed workflow. Give agents different knowledge, capabilities, and priorities. They must determine which collaborators can help and revise their plan when circumstances change. Running an existing discovery or logistics demo alone is not sufficient; demonstrate your own adaptation scenario and decision logic.

Example projects A coffee exchange finds alternative suppliers after a shipment fails; a disaster response team reallocates resources as conditions change; or a campus planning system adjusts when transportation or facilities become unavailable.

Success criteria Demonstrate dynamic discovery or capability-based selection; respond to at least one unexpected change or failure; resolve conflicting recommendations toward a shared goal; and explain why the system selected particular agents.

Can your AI team adapt when the original plan no longer works?

Level 3 Build a Collective Intelligence

Mission Create autonomous agents with competing objectives that deliberate, negotiate, and attempt to reach consensus without a central decision-maker.

Use at least three autonomous agents with distinct objectives or perspectives. Give them a real conflict to work through, rather than scripted agreement. Agents should exchange evidence, challenge proposals, and revise their positions when new information arrives. Shared messaging or facilitation is allowed, but no supervisor may impose the final decision.

Example projects Energy management agents balance cost, sustainability, and reliability; emergency response agents negotiate scarce resources; research agents evaluate conflicting hypotheses; or business agents negotiate investment across departments.

Success criteria Show genuine disagreement, a meaningful deliberation or negotiation mechanism, and the effect of new information. Compare the collective outcome with an individual-agent or simpler baseline using evidence from your scenario.

Consensus is not always the correct outcome. A successful system may recognize insufficient evidence, irreconcilable objectives, or an unacceptable compromise. Explain when it should stop, preserve disagreement, or seek human input.

Can independent AI agents make better decisions together than they can individually?

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How Your Project Will Stand out

Functionality Does the system work and accomplish its intended objective?

Innovation Is the approach creative or technically interesting?

Agent Intelligence Do agents demonstrate meaningful reasoning, autonomy, collaboration, or collective decision-making?

Engineering Is the solution well-designed, understandable, and technically sound?

Demonstration Can the team clearly explain and demonstrate its solution?

Learning and Evidence What did the team discover, and what evidence supports its conclusions?

Suggestions on What to Submit

1. Working Demonstration
2. Architecture Overview
3. Technical Implementation
4. Results and Lessons Learned

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Getting Started

You do not need to build a multi-agent system from scratch. AGNTCY is an open-source Linux Foundation project, originally launched by Cisco and collaborators, that provides infrastructure for agent discovery, identity, messaging, and observability across frameworks and systems [1, 2].

coffeeAGNTCY is one practical starting point. Its fictional coffee enterprise demonstrates interactions among agents through the Corto and Lungo applications. Corto provides a smaller entry point; Lungo explores richer coffee supply-chain interactions [3, 4]. These applications illustrate infrastructure and integration patterns; they do not by themselves satisfy all challenge levels.

Explore Run coffeeAGNTCY or your preferred framework. Observe the agents, messages, tools, and decisions before making changes.

Extend Add useful capabilities and responsibilities. Make clear which behavior is yours and which comes from the reference application.

Experiment Introduce a controlled change, conflicting recommendation, or competing objective appropriate to your level. Capture the resulting interactions.

Demonstrate Use a repeatable scenario. Show the outcome and explain why the agents acted as they did, including failures and unresolved disagreements.

For the AGNTCY path, start with the hands-on tutorial and official getting-started guide [3, 4], then use the reference application repository [2]. The app-sdk is a reference integration and learning layer used by coffeeAGNTCY; consult component-specific documentation for the current subsystem interfaces [5].

Beyond the Hackathon The Internet of Cognition

Cisco Outshift's Internet of Cognition (IoC) work explores how distributed agents and humans can share knowledge and intent and reason collectively. It extends the question from connecting agents to enabling shared understanding and collective decisions [9].

AGNTCY supplies interoperability building blocks. IoC provides a broader research context for exploring collective intelligence. This challenge connects the two through experiments in collaboration, adaptation, and disagreement; implementing a particular IoC protocol or product is not required.

In a coffee exchange, agents can first coordinate an order, then adapt when a supplier becomes unavailable, and finally negotiate competing goals such as price, delivery, and sustainability. The useful result is evidence of better decision-making, including when agreement should be withheld.

What happens when intelligence becomes a property of the group, as well as the individual?

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References and Learning Resources

Start with resources 2–4 to build. Use the video resources for orientation and the current documentation for setup. Resources reviewed October 7, 2026.

1. AGNTCY project and repositories

Official project overview and links to component source code.

https://github.com/agntcy

2. coffeeAGNTCY code repository

Reference application source, README, and application-specific setup instructions.

https://github.com/agntcy/coffeeAgntcy

3. Exploring CoffeeAGNTCY hands-on tutorial

Guided exercises for running and exploring the reference applications.

https://github.com/agntcy/coffeeAgntcy/blob/main/TUTORIAL.md

4. CoffeeAGNTCY official getting-started guide

Overview of Corto and Lungo with links to deployment and component guides.

https://docs.agntcy.org/coffee-agntcy/get-started/

5. AGNTCY reference Application SDK

Code and integration examples used by coffeeAGNTCY.

https://github.com/agntcy/app-sdk

6. AGNTCY official video channel and Briefs

Project-linked video resources for concepts and demonstrations.

https://www.youtube.com/@agntcy-lf

7. AGNTCY Briefs playlist

Short project introductions linked from the official AGNTCY repository.

https://www.youtube.com/playlist?list=PL49BrgsjXg5qVeRVqlX9O74W02q3c8fow

8. AGNTCY getting-started video tutorials

Video playlist recommended in the AGNTCY hackathon resource guide by Sujata Tibrewala.

https://www.youtube.com/playlist?list=PL49BrgsjXg5oy1erXapxA1ELkFIrE2dPl

9. Internet of Cognition

Cisco Outshift's research context and whitepaper resources.

https://outshift.cisco.com/internet-of-cognition

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